WHAT IS THE CLOUD? (January 2011)

…and what could we do?

OCHA

Definition[1]. In general, there is often confusion as to what the term "Cloud Computing" actually means. Perhaps the easiest way is to consider it a utility service for IT resources – when you need a new server or more disk space, it is simply available just as electricity is when you plug in an electronic device. When it is no longer needed, the IT resource is automatically removed. You only pay for what you use. With such resources, the cloud provides amazing opportunities on how OCHA could manage its information.

Classifying the Services in the Cloud. The services in the cloud can be classified into three general categories. These categories are:

1. Software-as-a-Service (SaaS)

- Complete package provided with access through the browser
- Latest updates are received transparently
- No need to install software
- Limited ability to customize
- No in-house administration
- Popular options include Google Docs and Salesforce.com (CRM software)

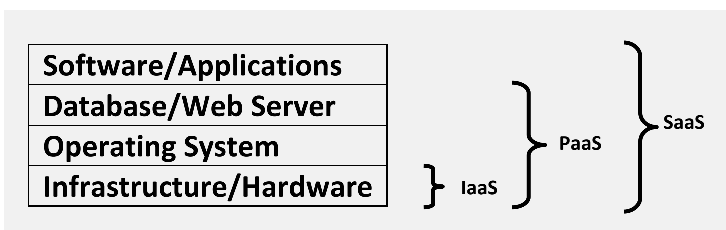
2. Platform-as-a-Service (PaaS)

- Computing platform (e.g. web server) with development software is provided
- Org concentrates on development as no need to maintain server and platform resources
- Cost efficient, scalable & easy use
- Popular options include Microsoft Azure and Google App Engine

3. Infrastucture-as-a-Service (IaaS)

- Infrastructure (e.g. physical servers) are delivered as a service
- Based on virtualization of machines
- Key feature is instant provisioning
- Major options are:
  - Amazon EC2
  - Eucalyptus
  - GoGrid
  - RackSpace

Layering the Cloud Services. In an attempt to better understand how these services relate to each other, they can be visualized with IaaS on the very bottom layer as it provides the basic (infrastructure and hardware) services. SaaS requires all layers to be in place to deliver a software service.



Cloud Models. Although the most commonly known and referenced type of cloud is the *public* cloud, there are actually four different cloud types. Each offer their own advantages and disadvantages.

1. Public

- Available to general public
- Managed by third party
- Pro: pay only for what is used
- Pro: optimized resource usage
- Pro: Easy trial and evaluations
- Con: Lack of total control of security, governance & reliability
- Con: compromise may be required to meet needs

2. Private

- Dedicated to single org
- Exist on or off-premise
- Pro: minimized pitfalls of public cloud
- Pro: easier to comply with corporate policies
- Pro: can be outsourced
- Con: requires expertise to configure and maintain (if self hosted)

3. Community

- Infrastructure and services shared by several orgs (e.g. humanitarian responders)
- Managed by orgs or a 3rd party
- Pro: Cost savings due to sharing
- Pro: Easier to implement standards
- Pro: leverage resources and services of multiple orgs

3. Hybrid

- Mixture of other cloud models
  - E.g. use public cloud for testing apps, but a private cloud to host core/approved apps
- Pro: Exploit features of different clouds to meet needs
- Pro: cloud bursting (use public cloud to handle overspill in demand for private cloud)

[1] Definition from Learning Tree's Course 1200: *Cloud Computing Technologies: A Comprehensive Hands-On Introduction*: The ability to add and remove transparently scalable IT resources (both hardware and software) from a seemingly infinite pool of resources in a self-service manner.

Cloud Computing Benefits. The benefits of cloud computing can be categorized into two groups: technical and business.

> The technical benefits are **evolutionary** in nature. Essentially, the cloud is a combination and integration of a number of existing technologies (server farms, high-speed networks, virtualization, etc). It reduces administration and maintenance tasks by potentially significant amounts as fewer on-premise installations are required, automatic redundancy is provided, versions are updated seamlessly, and resource requirement can be instantly matched based on changing demands.
>
> The business benefits are **revolutionary** in nature. The cloud disrupts traditional IT infrastructure and purchasing models as well as provides new ways to manage information. It introduces a seemingly infinite pool of computing resources that are available on demand and payable based on usage. More concretely, it provides financial benefits (lower admin and capital), agility (responsive, easier to share and collaborate, easier to change with new IM trends), operational benefits (opens innovative data processing and information sharing opportunities, reduces assets to track, requires less procurement and disposal), and flexibility.

Is the cloud for OCHA? Many IM colleagues are already making use of SaaS by using Google Apps. IMOs have made use of several SaaS components during emergencies for both working internally and collaborating with the volunteer tech community (VTC). However, there are significant potentials beyond SaaS. The cloud enables functionality, such as micro-tasking, that could revolutionize the way OCHA and the humanitarian community manage information. HumanitarianResponse.info could scale based on demand and we would only pay for what is used. By running our test servers, such as the OCHANet Staging server, on Amazon's EC2, we would only pay for the time that we actually use the server [i.e. about 8-16 cents/hour] rather than for our own hardware, electricity, maintenance, security, etc, etc. And, imagine if we wanted to test a new environment or server - we could simply "spin it up" on Amazon's EC2, evaluate it, and then decide if we like the results. The costs would be minimal, but the benefits enormous.

What Could We Do Next? Before OCHA can start using the cloud, it must change its mindset about traditional business models and agree that we are going to openly and optimistically test the cloud, accept failures, and be committed to pursuing successes. Possible steps for OCHA to consider are:

1. Educate OCHA staff (especially IM and IT) on the basics of the cloud by hiring an expert consultant, for a period of a few days, to provide a general introduction thereby ensuring everyone understands the overall concepts and to provide more in-depth knowledge to OCHA staff in the various cloud services and models.
2. Define and quickly deploy non-mission critical pilots in all three cloud service levels (SaaS, PaaS, and IaaS). Pilots could include Google Docs and Live Docs, but we should be preparing pilots in the other service levels such as the OCHANet Staging server (IaaS).
3. Evaluate the pilots as they progress so that we can quickly decide if such a solution is appropriate for OCHA. If so, figure out how to implement it as quickly as possible [i.e. we could be figuring out the red-tape before the pilot ends]. If not, define why and develop more appropriate pilots.
4. Work with our partners [e.g. IASC Task Force on IM] to identify possible areas of community cloud use for collaboration in preparedness and emergency response.
5. Continue to trial and train on the cloud to ensure that OCHA keeps itself modern and relevant.

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11 January 2011